

SPRINT 1 — BENCHMARKING PROTOCOL

Baseline Query Definitions: PostgreSQL vs InfluxDB

Team SG04 | Topic M5 — Human Activity Anomaly Detection | UCI HAR Dataset

1. Purpose & Scope

This document defines the five benchmark queries to be executed in Sprint 2, comparing PostgreSQL (relational) against InfluxDB (time-series). Each query targets a specific operation type and produces a measurable metric.

Queries are based on the UCI HAR dataset: 30 subjects, 9 raw signal channels at 50 Hz, 10,299 sliding windows of 2.56 seconds, and 561 engineered features per window.

2. Benchmark Queries

Q1 Range Scan — All Windows for Subject 1 [Range Scan]	
Sprint 2 Metric: Query latency in milliseconds	
PostgreSQL	InfluxDB (Flux)
<pre>SELECT r.recording_id, r.activity_id, r.window_duration_s, fv.feature_id, fv.value FROM recordings r JOIN feature_vectors fv ON r.recording_id = fv.recording_id WHERE r.subject_id = 1;</pre>	<pre>from(bucket: "har") > range(start: 2024-01-01T00:00:00Z) > filter(fn: (r) => r._measurement == "har_kinematic_signals" and r.subject_id == "1")</pre>
Q2 Aggregation — Mean Feature Value per Activity [Aggregation]	
Sprint 2 Metric: Query execution time in milliseconds	
PostgreSQL	InfluxDB (Flux)
<pre>SELECT a.activity_label, AVG(fv.value) AS mean_value FROM feature_vectors fv JOIN recordings r ON fv.recording_id = r.recording_id JOIN activities a ON r.activity_id = a.activity_id JOIN features f ON fv.feature_id = f.feature_id WHERE f.feature_name = 'tBodyAcc-mean()-X' GROUP BY a.activity_label;</pre>	<pre>from(bucket: "har") > range(start: 2024-01-01T00:00:00Z) > filter(fn: (r) => r._measurement == "har_windowed_features" and r._field == "f_tBodyAcc_mean_X") > group(columns: ["activity"]) > mean()</pre>
Q3 Write Test — Bulk Data Insertion [Write]	

Sprint 2 Metric: *Insertion throughput in rows per second*

PostgreSQL	InfluxDB (Flux)
-- 10,299 windows x 561 features -- = ~5.77 million rows INSERT INTO feature_vectors (recording_id, feature_id, value, created_at) VALUES (?, ?, ?, NOW()), ...;	// Python influxdb-client // write_api.write(// bucket="har", // record=data_frame //) // 10,299 pts x 9 channels // timed via write_api

Q4 Downsampling — Resample to 1-Second Bins [Downsample]

Sprint 2 Metric: *Query execution time in milliseconds*

PostgreSQL	InfluxDB (Flux)
SELECT r.subject_id, DATE_TRUNC('second', s.created_at) AS second_bucket, AVG(s.samples[1]) AS avg_acc_x FROM signals s JOIN recordings r ON s.recording_id = r.recording_id WHERE s.channel = 'body_acc_x' GROUP BY r.subject_id, second_bucket;	from(bucket: "har") > range(start: 2024-01-01T00:00:00Z) > filter(fn: (r) => r._measurement == "har_kinematic_signals" and r._field == "tBodyAcc_x") > aggregateWindow(every: 1s, fn: mean)

Q5 Count — Window Distribution per Activity [Count]

Sprint 2 Metric: *Query execution time in milliseconds*

PostgreSQL	InfluxDB (Flux)
SELECT a.activity_label, COUNT(r.recording_id) AS nb_windows FROM recordings r JOIN activities a ON r.activity_id = a.activity_id GROUP BY a.activity_label ORDER BY nb_windows DESC;	from(bucket: "har") > range(start: 2024-01-01T00:00:00Z) > filter(fn: (r) => r._measurement == "har_windowed_features") > group(columns: ["activity"]) > count()

3. Summary Table

#	Query	Type	Metric	Tables / Measurements
Q1	All windows — Subject 1	Range Scan	Latency (ms)	recordings + feature_vectors har_kinematic_signals
Q2	Mean feature per activity	Aggregation	Exec time (ms)	feature_vectors + activities har_windowed_features
Q3	Bulk data insertion	Write	Rows / sec	feature_vectors

				har_kinematic_signals
Q4	Downsample to 1-sec bins	Downsample	Exec time (ms)	signals har_kinematic_signals
Q5	Window count per activity	Count	Exec time (ms)	recordings + activities har_windowed_features

4. Metrics to Record in Sprint 2

For each query, the following metrics will be measured in both PostgreSQL and InfluxDB:

Metric	Unit	Description
Write Throughput	rows/sec	Rows inserted per second during bulk load (Q3)
Read Latency	ms	Time from query submission to first result (Q1)
Aggregation Time	ms	Total execution time for GROUP BY operations (Q2, Q5)
Downsample Time	ms	Execution time for 1-second binning over all data (Q4)
Disk Usage	MB	Storage consumed after full dataset ingestion